

Seqlocks (Sequence Locks)

1. What is Seqlock?

- Seqlock is a Linux synchronization mechanism (Linux 2.6)
- Used for **fast reader access + controlled writer updates**
- Solves **writer starvation problem in RW locks**

2. Core Idea

- Writers never wait for readers
- Readers never lock
- Readers verify data using a **sequence counter**
- If writer interferes → reader retries

3. Sequence Counter Rule

- Even number → data is stable
- Odd number → write in progress

4. Writer Behavior

- Acquire spinlock (only one writer)
- Increment seq → odd (start write)
- Update shared data
- Increment seq → even (end write)
- Release spinlock

5. Reader Behavior

- Read sequence before data
- Read shared data
- Read sequence again after data
- Compare both values
- If mismatch → retry

6. Reader Validation

- If `seq_before == seq_after AND even` → valid data
- Else → retry loop

7. Key Properties

- Readers are lock-free
- Writers are serialized (spinlock)
- Writers never blocked by readers
- Readers may retry multiple times

8. Restrictions

- Not suitable for pointer-based data
- Not for complex data structures
- Unsafe if data can change memory references

9. Reader vs Writer Summary

- Readers → fast, lock-free, retry-based
- Writers → exclusive, spinlock protected
- Writers have priority over readers

10. Advantages

- Very fast read performance
- No writer starvation
- Simple mechanism

11. Disadvantages

- Readers may retry repeatedly
- Not suitable for complex data
- No strict consistency guarantee during read

12. Use Cases

- System time (jiffies / wall clock)
- Kernel statistics
- Frequently read, rarely updated data

13. When to Use

- Many readers, few writers
- Small data size
- High-speed read requirement
- Writer starvation must be avoided

14. Key Takeaway

- Seqlock = **optimistic locking**
- Readers assume data is correct, verify later
- Writers always get immediate access

```
vboxuser@ubuntu2204:~/synchronizaation/SEQ_LOCKS$ ls
Makefile      README.md      seqlock_demo.mod  seqlock_demo.o
modules.order  seqlock_demo.c  seqlock_demo.mod.c
Module.symvers seqlock_demo.ko seqlock_demo.mod.o
vboxuser@ubuntu2204:~/synchronizaation/SEQ_LOCKS$ sudo dmesg | tail
[ 516.963456] Reader-2 : shared_data = 21
[ 517.668638] Writer : Updated shared_data = 22
[ 517.802374] Reader-1 : shared_data = 22
[ 517.987806] Reader-2 : shared_data = 22
[ 518.825260] Reader-1 : shared_data = 22
[ 519.009781] Reader-2 : shared_data = 22
[ 519.842872] Reader-1 : shared_data = 22
[ 520.033294] Reader-2 : shared_data = 22
[ 520.866047] Reader-1 : shared_data = 22
[ 521.057257] Reader-2 : shared_data = 22
• vboxuser@ubuntu2204:~/synchronizaation/SEQ_LOCKS$ sudo rmmod rcu demo
```